

RESEARCH PROFILE

My research studies the **structure, computation, and learning of values, equilibria, and strategies in stochastic games**, in which the same players interact again and again and present choices shape future interaction. During my Ph.D., I have studied four classes of such games: concurrent stochastic games, turn-based stochastic games, partially observable Markov decision processes, and robust Markov decision processes. I investigate these games from three perspectives:

- **Mathematical structure:** existence and characterization of values, equilibria, and optimal strategies, and the algebraic structure, memory, and numerical precision they require.
- **Computation:** the complexity of these problems, efficient algorithms, and the guarantees and limits of classical methods such as policy and strategy iteration.
- **Learning:** what can be learned about values and strategies from finitely many observations, and how robust models account for estimation error.

EDUCATION

- **Ph.D. in Computer Science** 2021 – Present
Institute of Science and Technology Austria (ISTA), Klosterneuburg, Austria
- **B. Sc. in Computer Engineering** 2016 – 2021
Sharif University of Technology, Tehran, Iran GPA: 18.93/20

PUBLICATIONS

- “On the Complexity of Discounted Robust MDPs with L_p Uncertainty Sets”
Ali Asadi, Krishnendu Chatterjee, Alipasha Montaseri, Ali Shafiee
in the *40th Annual Conference on Neural Information Processing Systems (NeurIPS’26, Spotlight)*, 2026.
- “Generalized Bidding Games: Where Bidding and Stochastic Games Meet”
Ali Asadi, Thomas A. Henzinger, Ehsan Kafshdar Goharshady, Pavol Kebis, Kaushik Mallik
in the *Proceedings of the 37th International Conference on Concurrency Theory (CONCUR’26)*, 2026.
- “PAC Learning in Turn-Based Stochastic Games with Reachability Objectives: A Decentralized Private Approach via Expected Conditional Distance”
Ali Asadi, Krishnendu Chatterjee, Pavol Kebis
in the *Proceedings of the 37th International Conference on Concurrency Theory (CONCUR’26)*, 2026.
- “Strongly Polynomial Time Complexity of Policy Iteration for L_∞ Robust MDPs”
Ali Asadi, Krishnendu Chatterjee, Ehsan Goharshady, Mehrdad Karrabi, Alipasha Montaseri, Carlo Pagano
in the *Proceedings of the 39th Annual Conference on Learning Theory (COLT’26)*, PMLR 336:427–457, 2026.
- “Qualitative Analysis of ω -Regular Objectives on Robust MDPs”
Ali Asadi, Krishnendu Chatterjee, Ehsan Goharshady, Mehrdad Karrabi, Ali Shafiee
in *The 40th Annual AAAI Conference on Artificial Intelligence (AAAI’26)*.
- “Revealing POMDPs: Qualitative and Quantitative Analysis for Parity Objectives”
Ali Asadi, Krishnendu Chatterjee, David Lurie, Raimundo Saona
in *The 40th Annual AAAI Conference on Artificial Intelligence (AAAI’26)*.
- “ ϵ -Stationary Nash Equilibria in Multi-player Stochastic Graph Games”
Ali Asadi, Léonard Brice, Krishnendu Chatterjee, K. S. Thejaswini
in the *45th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS’25)*.

- “Limit-sure reachability for small memory policies in POMDPs is NP-complete”
Ali Asadi, Krishnendu Chatterjee, Raimundo Saona, Ali Shafiee
in the 41st Conference on Uncertainty in Artificial Intelligence (**UAI’25**).
- “Lower Bound on Howard Policy Iteration for Deterministic Markov Decision Processes”
Ali Asadi, Krishnendu Chatterjee, Jakob de Raaij
in the 41st Conference on Uncertainty in Artificial Intelligence (**UAI’25**).
- “Concurrent Stochastic Games with Stateful-discounted and Parity Objectives: Complexity and Algorithms”
Ali Asadi, Krishnendu Chatterjee, Raimundo Saona, Jakub Svoboda
in the 44th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (**FSTTCS’24**).
- “Deterministic Sub-exponential Algorithm for Discounted-sum Games with Unary Weights”
Ali Asadi, Krishnendu Chatterjee, Raimundo Saona, Jakub Svoboda
in the Proceedings of the 39th Annual Symposium on Logic in Computer Science (**LICS’24**), 2024.
- “Polynomial reachability witnesses via Stellensätze”
Ali Asadi, Krishnendu Chatterjee, Hongfei Fu, Amir Kafshdar Goharshady, Mohammad Mahdavi
in the Proceedings of the 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation (**PLDI’21**), 2021.
- “Faster Algorithms for Quantitative Analysis of MCs and MDPs with Small Treewidth”
Ali Asadi, Krishnendu Chatterjee, Amir Kafshdar Goharshady, Kiarash Mohammadi, Andreas Pavlogiannis
in the Proceedings of the 18th International Symposium on Automated Technology for Verification and Analysis (**ATVA’20**), 2020.

MANUSCRIPTS UNDER REVIEW

- “The Complexity of Approximating the Value in Revealing POMDPs with Long-Run Average Objectives”
Ali Asadi, Krishnendu Chatterjee, David Lurie
submitted to *Mathematics of Operations Research* (**MOOR**).
- “Algorithms with Smoothed Polynomial-Time Complexity for Deterministic Discounted-sum and Mean-payoff Games”
Ali Asadi, Krishnendu Chatterjee, Ruichen Luo
submitted to the 38th Annual ACM-SIAM Symposium on Discrete Algorithms (**SODA’27**).
- “Values of Absorbing Recursive Games Express All Real Algebraic Numbers”
Ali Asadi, Krishnendu Chatterjee
submitted to *Mathematics of Operations Research* (**MOOR**).

TALKS AND POSTERS

- “Revealing POMDPs: Qualitative and Quantitative Analysis for Parity Objectives”
Talk: Highlights of Logic, Games and Automata (Highlights’26), Vienna, Austria Sept. 2026
Poster: Paris Game Theory Festival 2026, Paris, France June 2026
Poster: AAAI Conference on Artificial Intelligence (AAAI’26), Singapore Jan. 2026
Seminar talk: Combinatorics, Games and Optimisation Seminar, LSE, London, UK Dec. 2025
- “Limit-sure reachability for small memory policies in POMDPs is NP-complete”
Poster: 30 Years of Game Theory at Institut Henri Poincaré, Paris, France Oct. 2025
- “Deterministic Sub-exponential Algorithm for Discounted-sum Games with Unary Weights”
Talk: International Conference on Reachability Problems (RP’24), Vienna, Austria Sept. 2024
Talk: ACM/IEEE Symposium on Logic in Computer Science (LICS’24), Tallinn, Estonia July 2024
Poster: Mediterranean Game Theory Symposium 2024, Marseille, France May 2024

HONORS AND AWARDS

- **Silver medal** in the 10th Asia-Pacific Informatics Olympiad (APIO) 2016
- **Gold and Silver medals** in the 25th and 24th Iranian National Olympiad in Informatics (INOI) 2015, 2014
- **2nd place** in the ACM-ICPC West Asia Regional Contest, Tehran, Iran. 2016
- Iranian National Elites Foundation grant for undergraduate studies 2014 – 2021
- Research Scholarship from the Austrian Exchange Agency (OeAD-GmbH) 2019

SCIENTIFIC SERVICE

- **Program Committee Member**
 - [AAAI Conference on Artificial Intelligence \(AAAI'27\)](#) 2027
 - [AAAI Conference on Artificial Intelligence \(AAAI'26\)](#) 2026
- **Journal Reviewer**
 - [SIAM Journal on Control and Optimization \(SICON\)](#) 2026
 - [Information and Computation](#) 2026
- **Conference Reviewer**
 - [Conference on Uncertainty in Artificial Intelligence \(UAI'26\)](#) 2026
 - [Foundations of Software Technology and Theoretical Computer Science \(FSTTCS'25\)](#) 2025
 - [International Symposium on Mathematical Foundations of Computer Science \(MFCS'25\)](#) 2025
 - [International Colloquium on Automata, Languages, and Programming \(ICALP'24\)](#) 2024

TEACHING EXPERIENCE

University Teaching

- **Teaching Assistant, Institute of Science and Technology Austria (ISTA)** 2024 – 2025
Held office hours and prepared and graded homework assignments.
 - Computer Science Track Core Course Spring 2024; Spring 2025
 - Formal Methods: Algorithmic Approaches Spring 2024
- **Teaching Assistant, Sharif University of Technology** 2017 – 2018
Prepared and graded homework assignments.
 - Data Structures and Algorithms Spring 2018
 - Probability and Statistics Fall 2017

Olympiad Teaching and Educational Service

- **Lecturer, Algorithms and Data Structures** 2016 – 2019
Prepared students for the Iranian National Olympiad in Informatics (INOI).
 - Allameh Helli High School 2016 – 2018
 - Allameh Tabatabaee High School 2016 – 2019
- **Scientific Committee Member and Problem Author, INOI**
 - INOI Final Exams 2022 – 2025
 - IOI Team Selection Exams 2017
 - INOI Summer Camp Programming Exams 2016

RESEARCH SUPERVISION

Research Interns and Rotation Students, Institute of Science and Technology Austria (ISTA)

- **Pavol Kebis** Fall 2025
 - “PAC Learning in Turn-Based Stochastic Games with Reachability Objectives: A Decentralized Private Approach via Expected Conditional Distance” (**CONCUR’26**)
- **Jack Quirion** Summer 2025
 - “Multi-Objective Reachability in Stochastic Games” (**planned submission to ICALP’27**)
- **Alipasha Montaseri** Summer 2025
 - “Strongly Polynomial Time Complexity of Policy Iteration for L_∞ Robust MDPs” (**COLT’26**)
 - “On the Complexity of Discounted Robust MDPs with L_p Uncertainty Sets” (**NeurIPS’26, Spotlight**)
- **Ali Shafiee** Summer 2024
 - “Qualitative Analysis of ω -Regular Objectives on Robust MDPs” (**AAAI’26**)
 - “Limit-sure reachability for small memory policies in POMDPs is NP-complete” (**UAI’25**)
- **Jakob de Raaij** Summer 2023
 - “Lower Bound on Howard Policy Iteration for Deterministic Markov Decision Processes” (**UAI’25**)

INDUSTRY EXPERIENCE

- **Engineering Manager, Divar Co.** Sep. 2019 – Sep. 2021
Led a team of 10 engineers building scalable advertising infrastructure.
- **Software Engineer, CafeBazaar Co.** July 2017 – July 2019
Developed high-scale payment and deployment systems.